Your Brain vs AI

How to stay sharp in the age of AI

Manish Kumar Singh

A 10-second test

Without looking: what is a close friend's phone number?

Now: where would you find it?

Most of us know where, not what.

When we expect to look something up later, we remember it less, but we remember where it is.

Source: Sparrow, Liu & Wegner (2011), Google effects on memory, Science 333:776-778

What we will cover

1. How attention works
2. How memory works
3. What leaning on AI does to thinking: 3 studies, stated honestly
4. Five habits that protect focus and memory
5. Your questions

Your brain, in numbers

About 86 billion neurons

About 2% of body weight, about 20% of the body's energy: roughly the power of a dim bulb

It changes with use: London taxi drivers who learned the city's streets had a larger back part of the hippocampus, the brain's map-maker

Sources: Azevedo et al. (2009), J Comp Neurol 513:532-541; Raichle & Gusnard (2002), PNAS 99:10237-10239; Maguire et al. (2000), PNAS 97:4398-4403

Attention is a spotlight, not a floodlight

Working memory holds only about 4 chunks at once

Switching tasks leaves 'attention residue': part of your mind stays on the last task

So every switch has a cost, even a quick check of a message

Sources: Cowan (2001), Behavioral and Brain Sciences 24:87-114; Leroy (2009), Organizational Behavior and Human Decision Processes 109:168-181

Screens and the shrinking spotlight

Researchers logged how long people stay on one screen before switching

Around 2004: about 2.5 minutes

In recent years: about 47 seconds on average

Fast switching was linked with higher stress in her studies

Source: Gloria Mark (2023), Attention Span, Hanover Square Press (summary of her field studies at UC Irvine)

How memory actually works

Encode: pay attention and make meaning

Consolidate: sleep replays and strengthens new memories

Retrieve: every time you recall something, it gets easier to recall again

Without review, we forget much of new material within days: the forgetting curve

Sources: Diekelmann & Born (2010), Nature Reviews Neuroscience 11:114-126; Murre & Dros (2015), PLOS ONE 10:e0120644 (replication of Ebbinghaus 1885)

Recalling beats re-reading

Students read a text, then either re-read it or tested themselves

Five minutes later, re-readers did slightly better

One week later, the self-testers remembered clearly more

Effort now is memory later

Source: Roediger & Karpicke (2006), Psychological Science 17:249-255

What does leaning on AI do to thinking?

Three studies, with their limits

Study 1: students and maths

About 1,000 high-school students in Turkey, real classes

Plain ChatGPT access raised practice scores

When the AI was removed for the exam, those students scored 17% lower than students who never had it

An AI tutor that gave hints, not answers, largely avoided that harm

Source: Bastani et al. (2025), Generative AI without guardrails can harm learning, PNAS 122:e2422633122

Study 2: brains while writing essays

54 adults wrote essays with ChatGPT, with a search engine, or with no tools, while EEG recorded brain activity

The ChatGPT group showed the weakest brain connectivity of the three

Many in that group struggled to quote from the essay they had just written

Caution: small study, published as a preprint

Source: Kosmyna et al. (2025), Your Brain on ChatGPT: Accumulation of Cognitive Debt, MIT Media Lab, arXiv:2506.08872

Study 3: people at work

319 knowledge workers described 936 real uses of AI at work

More trust in the AI went with less critical thinking; more self-confidence went with more

A separate survey of 666 people linked heavier AI use with lower critical-thinking scores, via 'offloading'

These are surveys: they show links, not cause and effect

Sources: Lee et al. (2025), CHI 2025, Microsoft Research and Carnegie Mellon University; Gerlich (2025), Societies 15(1):6

The honest summary

AI is a powerful tool. The risk is not using it; it is letting it do the thinking

Use AI after you have thought, not instead of thinking

Make it ask you questions, not just give you answers

Keep the skills you want to keep in regular use

Based on: Bastani et al. (2025); Kosmyna et al. (2025); Lee et al. (2025)

Habits 1 and 2

1. Think first, then ask AI. Write your own answer or outline in 2 minutes, then compare

2. Test yourself instead of re-reading. Close the book, write what you remember, then check

Ask AI to quiz you, not to summarise for you

Sources: Bastani et al. (2025), PNAS; Roediger & Karpicke (2006), Psychological Science

Habits 3 and 4

3. Space it out: short reviews spread over days beat one long cram

4. One task at a time: phone out of reach, one tab, notifications off for a 25 to 50 minute block

End each block by noting where you stopped: it reduces attention residue

Sources: Cepeda et al. (2006), Psychological Bulletin 132:354-380; Leroy (2009), OBHDP; Mark (2023)

Habit 5: sleep and move

Sleep is when the brain locks in what you learned. An all-nighter trades memory for hours

Regular aerobic exercise: in a one-year trial, older adults who walked briskly grew their hippocampus by about 2%

A walk before study or deep work is a cheap upgrade

Sources: Diekelmann & Born (2010), Nat Rev Neurosci; Erickson et al. (2011), PNAS 108:3017-3022

Try this for 7 days, then questions

Day 1-7: think first, then ask AI

Each evening: 5 minutes of recall on what you learned today

One phone-free deep-work block a day

Notice what changes. Then over to you: what would you like to ask?

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